

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

TASK 1 REPORT

Enhanced House Price Prediction Using Linear Regression

Submitted By:

Ishpreet Kaur

Tool Used:

Google Colab (Python)

Technology Stack:

Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn

ABSTRACT

Machine Learning is used to build predictive models from historical data. This project develops a Linear Regression model using the California Housing Dataset and includes preprocessing, exploratory data analysis, feature engineering, prediction and evaluation.

1. INTRODUCTION

This project predicts house prices using Linear Regression and demonstrates preprocessing, visualization, model development and evaluation.

2. OBJECTIVES

- Understand ML workflow
- Apply Linear Regression
- Perform EDA
- Create engineered features
- Evaluate model performance

3. DATASET DESCRIPTION

Dataset: California Housing Dataset

Features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude

Target Variable: Price

4. FEATURE ENGINEERING

Additional feature created:

$\text{Rooms_per_House} = \text{Average Rooms} / (\text{House Age} + 1)$

Purpose: Improve analysis and presentation.

5. EXPLORATORY DATA ANALYSIS

Performed dataset inspection, missing value checking, statistical summary, distribution analysis and correlation heatmap.

6. MODEL DEVELOPMENT

Steps: Load Dataset → Preprocessing → Feature Engineering → Train/Test Split → Train Model → Prediction → Evaluation.

7. MODEL EVALUATION

Metrics Used: MAE, RMSE and R^2 Score. Additional analysis included prediction comparison and feature contribution.

8. RESULTS

Outputs generated included graphs, prediction comparison and actual vs predicted analysis.

9. CONCLUSION

The project successfully implemented Linear Regression using California Housing Dataset and demonstrated complete ML workflow.

REFERENCES

Scikit-learn Documentation
California Housing Dataset
Google Colab